

Gaussian Rigidity from Symmetrized Moment Generating Functions: Chi-Square Quadratic Forms and Sharp Reflection Stability

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Abstract

We prove a Gaussian rigidity principle based on the symmetrized moment generating function $M_X(s)M_X(-s)$. First, let $X = (X_i)$ have independent centered coordinates and let $A \succeq 0$. If $X^\top AX$ has a chi-square law and each active coordinate satisfies the local dominance $M_i(s)M_i(-s) \geq e^{\sigma_i^2 s^2}$, then every active coordinate is Gaussian. Second, under a fixed exponential-moment bound, the local estimate

$$\sup_{|s| \leq \rho} |\log M_X(s) + \log M_X(-s) - s^2| \leq \Delta$$

implies

$$d_K(X, \mathcal{N}(0, 1)) \leq C \sqrt{\frac{\log \log(1/\Delta)}{\log(1/\Delta)}}.$$

The order is optimal in general, as shown by a Laguerre construction. The proof recovers the factor from reflected MGF information through a growing zero-free region. We also derive exact and quantitative consequences for the sample variance and show that total-variation stability fails.

1 Introduction and main results

The exact chi-square law of the sample variance is one of the classical signatures of Gaussian samples. The unrestricted converse for a fixed sample size $n \geq 3$ remains a longstanding characterization problem. Existing results impose additional structure, including symmetry or infinite divisibility; see [2, 6, 8, 10–12]. We do not claim to resolve that unrestricted problem. The sample variance is a chi-square quadratic form, and spherical averaging of its Laplace transform naturally brings in the symmetrized quantity $M_X(s)M_X(-s)$.

For a centered variance-one random variable X , let $M_X(s) = \mathbb{E}e^{sX}$ whenever finite and define

$$R_X(s) := \log M_X(s) + \log M_X(-s) - s^2, \quad \Delta_\rho(X) := \sup_{|s| \leq \rho} |R_X(s)|. \quad (1)$$

We work under the fixed exponential envelope

$$\mathbb{E}e^{2\tau|X|} \leq K. \quad (2)$$

If X' is an independent copy of X , then $M_{X-X'}(s) = M_X(s)M_X(-s)$. The difficulty is that reflection removes the Fourier phase. Our stability theorem shows that the factor can nevertheless be recovered, at the optimal logarithmic scale, under (2).

The proof uses two scales. A fixed complex neighborhood yields high-order moment information for $X - X'$; after truncation, this produces a Gaussian product approximation on a disk of radius of order $\sqrt{m}$. Zero-freeness on this disk allows one to take a logarithm and recover the factor. The exact argument is separate and uses Jensen equality for the spherical transform.

Exact rigidity

The first result is the positive-semidefinite quadratic-form theorem that motivates the reflection condition.

Theorem 1.1 (Exact PSD rigidity). *Let $X = (X_1, \dots, X_d)$ have independent coordinates with $\mathbb{E}X_i = 0$ and $\text{Var}(X_i) = \sigma_i^2$. Let $A = A^\top \succeq 0$ have rank $r \geq 1$, and suppose*

$$X^\top A X \sim \chi_r^2.$$

For every active coordinate i with $A_{ii} > 0$, assume that $M_i(s) = \mathbb{E}e^{sX_i}$ is finite in a neighborhood of the origin and

$$M_i(s)M_i(-s) \geq e^{\sigma_i^2 s^2}$$

there. Then every active X_i is Gaussian, $X_i \sim \mathcal{N}(0, \sigma_i^2)$.

The dominance condition is neither a reformulation of symmetry nor of infinite divisibility. For an infinitely divisible centered law with finite variance, the Lévy–Khintchine representation gives, for every s such that both $M_X(s)$ and $M_X(-s)$ are finite,

$$\log M_X(s) + \log M_X(-s) - \sigma^2 s^2 = 2 \int \left(\cosh(sx) - 1 - \frac{s^2 x^2}{2} \right) \nu(dx) \geq 0.$$

The converse fails. If X takes the values 3 and $-1/3$ with probabilities $1/10$ and $9/10$, then it is centered and has variance one. It is not infinitely divisible: if a bounded non-degenerate law of support diameter d were an n -fold convolution of an i.i.d. root, that root would have support diameter at most d/n , and Popoviciu’s inequality would give $\text{Var}(X)/n \leq d^2/(4n^2)$, which is impossible for large n . Its fourth cumulant is $46/9$, so $R_X(s) = 23s^4/54 + O(s^6)$ and the local dominance holds. This example is asymmetric. In contrast, a Rademacher variable ε satisfies $R_\varepsilon(s) = 2 \log \cosh s - s^2 = -s^4/6 + O(s^6)$, which is negative for all sufficiently small nonzero s .

Sharp reflection stability

The second result recovers a factor from local reflected-MGF information without assuming the dominance sign.

Theorem 1.2 (Sharp reflection stability). *Fix $\rho, \tau, K > 0$. There exist constants $C < \infty$ and $\Delta_0 \in (0, e^{-e})$, depending only on (ρ, τ, K) , such that the following holds. If*

$$\mathbb{E}X = 0, \quad \mathbb{E}X^2 = 1, \quad \mathbb{E}e^{2\tau|X|} \leq K,$$

and

$$|R_X(s)| \leq \Delta, \quad |s| \leq \rho,$$

for some $0 < \Delta \leq \Delta_0$, then

$$d_K(X, \mathcal{N}(0, 1)) \leq C \sqrt{\frac{\log \log(1/\Delta)}{\log(1/\Delta)}}.$$

The order in this theorem cannot be improved uniformly in general. The obstruction is supplied by high-order odd perturbations whose first m odd moments vanish.

Theorem 1.3 (Laguerre lower bound). *There are constants $\rho, \tau, K, c, C > 0$ and centered variance-one random variables W_m such that the same uniform exponential-integrability and local dominance assumptions hold, $\Delta_m := \Delta_\rho(W_m) \downarrow 0$, and*

$$\log \frac{1}{\Delta_m} = 2m \log m + O(m), \quad d_K(W_m, \mathcal{N}(0, 1)) \geq cm^{-1/2}.$$

Consequently

$$d_K(W_m, \mathcal{N}(0, 1)) \geq c' \sqrt{\frac{\log \log(1/\Delta_m)}{\log(1/\Delta_m)}}$$

for all sufficiently large m .

2 Exact rigidity for chi-square quadratic forms

We prove Theorem 1.1 and then apply it to the ordinary sample variance. Write $A = B^\top B$, where $B \in \mathbb{R}^{r \times d}$ has full row rank, and let b_i be the i th column of B . If V is uniform on S^{r-1} , then spherical averaging gives

$$\mathbb{E}_V \prod_{i=1}^d M_i(t \langle b_i, V \rangle) = \mathbb{E}_{X, V} e^{t \langle BX, V \rangle} = \mathbb{E}_{\Psi_r} (t \sqrt{X^\top A X}). \quad (3)$$

Since $X^\top A X \sim \chi_r^2$, the final quantity in (3) equals

$$\mathbb{E}_{G, V} e^{t \langle G, V \rangle} = e^{t^2/2}, \quad G \sim \mathcal{N}(0, I_r). \quad (4)$$

All expectations below are legitimate for sufficiently small $|t|$, because the coordinate MGFs exist on a common neighborhood of the origin and the sphere is compact.

Define

$$C_i(s) := \log M_i(s) - \frac{\sigma_i^2 s^2}{2}$$

and

$$H_t(v) := \sum_{i=1}^d C_i(t \langle b_i, v \rangle) + \frac{t^2}{2} \left(\sum_{i=1}^d \sigma_i^2 \langle b_i, v \rangle^2 - 1 \right). \quad (5)$$

The quadratic terms cancel inside the exponential, so (4) implies

$$\mathbb{E}_V e^{H_t(V)} = e^{-t^2/2} \mathbb{E}_V \prod_i M_i(t \langle b_i, V \rangle) = 1.$$

Jensen's inequality therefore gives $\mathbb{E}_V H_t(V) \leq 0$.

The mean of the quadratic correction in (5) is zero. Indeed,

$$\mathbb{E}_V \sum_i \sigma_i^2 \langle b_i, V \rangle^2 = \frac{1}{r} \sum_i \sigma_i^2 \|b_i\|^2 = \frac{1}{r} \mathbb{E}(X^\top A X) = 1,$$

where the last equality uses the mean of χ_r^2 . Since V and $-V$ have the same law, coordinatewise dominance now gives

$$\begin{aligned} 2\mathbb{E}_V H_t(V) &= \sum_{i=1}^d \mathbb{E}_V [\log M_i(t\langle b_i, V \rangle) + \log M_i(-t\langle b_i, V \rangle) - \sigma_i^2 t^2 \langle b_i, V \rangle^2] \\ &= \sum_{i=1}^d \mathbb{E}_V R_i(t\langle b_i, V \rangle) \geq 0, \end{aligned}$$

where $R_i(s) = \log M_i(s) + \log M_i(-s) - \sigma_i^2 s^2$. Hence $\mathbb{E}_V H_t(V) = 0$, and every summand in the last display has zero expectation.

For an active coordinate, $A_{ii} = \|b_i\|^2 > 0$. If $r \geq 2$, the random variable $\langle b_i, V \rangle$ has a density that is strictly positive on the interior of $[-\|b_i\|, \|b_i\|]$. Continuity of R_i therefore implies $R_i(s) = 0$ on a neighborhood of the origin. If $r = 1$, then $V \in \{-1, 1\}$ and varying t directly gives the same conclusion, because R_i is even. Thus, for every active coordinate,

$$M_i(s)M_i(-s) = e^{\sigma_i^2 s^2}$$

near the origin. If $\sigma_i^2 = 0$, then $X_i = 0$ almost surely. If $\sigma_i^2 > 0$, an independent copy X'_i gives that $X_i - X'_i$ has the MGF of $\mathcal{N}(0, 2\sigma_i^2)$ near zero and hence has that Gaussian law. The Lévy–Cr  mer decomposition theorem applied to the independent sum $X_i + (-X'_i)$ then implies that X_i is Gaussian.

This proves Theorem 1.1. For Corollary 5.1, take

$$A = I_n - \frac{1}{n} \mathbf{1}\mathbf{1}^\top.$$

It is a rank- $n - 1$ orthogonal projection, every diagonal entry is $(n - 1)/n > 0$, and its quadratic form is exactly $\sum_i (X_i - \bar{X})^2$. The corollary follows immediately.

3 Sharp reflection stability

Throughout this section let X' be an independent copy of X , set

$$Y = X - X', \quad G \sim \mathcal{N}(0, 2),$$

and define

$$F(z) := M_Y(z) - e^{z^2}, \quad L := \log(1/\Delta).$$

We prove Theorem 1.2 through five lemmas.

3.1 High-order reflected moment transfer

Lemma 3.1 (Local defect to high-order moments). *Fix ρ, τ, K . For every $A > 0$ there is $c_0 = c_0(A, \rho, \tau, K) > 0$ such that, with*

$$m = \left\lfloor c_0 \frac{L}{\log L} \right\rfloor,$$

all sufficiently small Δ satisfy

$$\left| \mathbb{E}Y^k - \mathbb{E}G^k \right| \leq e^{-Am \log m}, \quad 0 \leq k \leq 2m.$$

In particular,

$$\mathbb{E}|Y|^{2m} \leq (Cm)^m.$$

Proof. Choose a fixed $0 < \rho_0 < \min\{\rho, \tau\}$ and a symmetric ellipse whose upper and lower half-ellipses lie in the strip $|\Re z| < 2\tau$. Exponential integrability gives a bound $|F(z)| \leq B_0$ on this ellipse, with B_0 depending only on (ρ, τ, K) . On the real diameter, the defect hypothesis gives

$$|F(s)| = e^{s^2} |e^{R_X(s)} - 1| \leq C_0 \Delta, \quad |s| \leq \rho_0,$$

where $F(s) = M_Y(s) - e^{s^2}$ and $Y = X - X'$. Apply the subharmonic two-constants theorem to $u(z) = \log |F(z)|$ on each half-ellipse. If $\omega(z)$ denotes the harmonic measure of the real diameter, then

$$u(z) \leq \omega(z) \log(C_0 \Delta) + (1 - \omega(z)) \log B_0.$$

The harmonic-measure lower bound on a fixed inner disk gives $\omega(z) \geq \omega_0 > 0$ whenever $|z| \leq r_0$, after possibly shrinking r_0 . Thus there are constants $r_0 > 0$, $\omega_0 > 0$, and $C_1 < \infty$ such that

$$\sup_{|z| \leq r_0} |F(z)| \leq C_1 \Delta^{\omega_0}.$$

The Cauchy estimate consequently gives

$$|F^{(k)}(0)| \leq C_1 k! r_0^{-k} \Delta^{\omega_0}.$$

For $k \leq 2m$ and $m = c_0 L / \log L$, Stirling's bound shows that the logarithm of the right-hand side is at most $-\omega_0 L + O(c_0 L)$. Choosing c_0 so that $(A + 3)c_0 < \omega_0$ gives $|F^{(k)}(0)| \leq e^{-Am \log m}$. Since $F^{(k)}(0) = \mathbb{E}Y^k - \mathbb{E}G^k$, the moment conclusion follows. Finally,

$$\mathbb{E}G^{2m} = \frac{(2m)!}{m!} \leq (Cm)^m,$$

and the negligible moment mismatch gives the stated $2m$ -th moment bound. $\square$

3.2 Tail lifting and bounded truncation

Lemma 3.2 (Square-root tail lifting). *There are constants $C, c > 0$ with the following property. For every sufficiently large fixed B , if $N = B\sqrt{m}$, then*

$$\mathbb{P}(|X| > N) \leq C e^{-\beta_B m},$$

where $\beta_B \rightarrow \infty$ as $B \rightarrow \infty$. If X_N denotes the conditional law $X \mid \{|X| \leq N\}$, then

$$d_{\text{TV}}(X, X_N) \leq C e^{-\beta_B m}.$$

Moreover

$$|\mathbb{E}X_N| + |\text{Var}(X_N) - 1| \leq C(1 + N^2)e^{-cN}.$$

Proof. Let a be a median of X . Since $\mathbb{E}X^2 = 1$, $|a| \leq \sqrt{2}$. The safe symmetrization estimate is

$$\mathbb{P}(|X - a| > x) \leq 4\mathbb{P}(|X - X'| > x).$$

For $N = B\sqrt{m} > \sqrt{2}$, Lemma 3.1 and Markov's inequality give

$$\mathbb{P}(|X| > N) \leq \mathbb{P}(|X - a| > N - \sqrt{2}) \leq 4 \frac{\mathbb{E}|Y|^{2m}}{(N - \sqrt{2})^{2m}} \leq C \left(\frac{C_0}{B^2} \right)^m.$$

Thus one may take $\beta_B = \frac{1}{2} \log(B^2/C_0)$ for sufficiently large B . If $p_N = \mathbb{P}(|X| > N)$, conditioning gives

$$d_{\text{TV}}(X, X_N) \leq p_N.$$

The exponential envelope also gives

$$\mathbb{E}(|X|\mathbf{1}_{\{|X|>N\}}) \leq Ce^{-cN}, \quad \mathbb{E}(X^2\mathbf{1}_{\{|X|>N\}}) \leq C(1+N^2)e^{-cN}.$$

Since $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$,

$$|\mathbb{E}X_N| \leq \frac{Ce^{-cN}}{1-p_N}, \quad |\mathbb{E}X_N^2 - 1| \leq \frac{p_N + C(1+N^2)e^{-cN}}{1-p_N}.$$

It follows that $|\mathbb{E}X_N| + |\text{Var}(X_N) - 1| \leq C(1+N^2)e^{-cN}$ after increasing C . $\square$

3.3 A growing real Gaussian-product window

Lemma 3.3 (Real-axis product approximation). *For B fixed sufficiently large, there is $a_0 = a_0(B, \rho, \tau, K) > 0$ such that, with*

$$R = a_0\sqrt{m}$$

and φ_N the characteristic function of X_N ,

$$\sup_{|t| \leq 2R} \left| \varphi_N(t)\varphi_N(-t) - e^{-t^2} \right| \leq e^{-\gamma_1 m}$$

for some $\gamma_1 > 0$.

Proof. Use the $2m-1$ Taylor polynomial of the characteristic function of $Y = X - X'$ and the real-variable remainder

$$\left| e^{iu} - \sum_{k=0}^{2m-1} \frac{(iu)^k}{k!} \right| \leq \frac{|u|^{2m}}{(2m)!}.$$

The comparison has three contributions:

$$\begin{aligned} |\varphi_Y(t) - e^{-t^2}| &\leq \frac{|t|^{2m}\mathbb{E}|Y|^{2m}}{(2m)!} + \frac{|t|^{2m}\mathbb{E}|G|^{2m}}{(2m)!} \\ &\quad + \sum_{k=0}^{2m-1} \frac{|t|^k}{k!} |\mathbb{E}(iY)^k - \mathbb{E}(iG)^k|, \end{aligned}$$

where $G \sim \mathcal{N}(0, 2)$. On $|t| \leq 2R$, the first two terms are bounded by $(Ca_0^2)^m$ after Stirling, while the third is super-exponentially smaller by Lemma 3.1. Choose a_0 small. Since $\varphi_Y(t) = \varphi_X(t)\varphi_X(-t)$ and the truncation error is bounded by Lemma 3.2, transfer to X_N gives

$$\varphi_N(t)\varphi_N(-t) = \varphi_{X_N - X'_N}(t) = e^{-t^2} + O(e^{-\gamma_1 m}).$$

$\square$

3.4 Complex propagation and a zero-free factor disk

Lemma 3.4 (Growing zero-free disk). *The constants B and a_0 can be chosen so that, for $R = a_0\sqrt{m}$,*

$$\varphi_N(z) \neq 0, \quad |z| \leq R.$$

More precisely, for some $\gamma_2 > a_0^2$,

$$\sup_{|z| \leq R} \left| \varphi_N(z)\varphi_N(-z) - e^{-z^2} \right| \leq e^{-\gamma_2 m}.$$

Proof. Since $|X_N| \leq N$, φ_N is entire and

$$|\varphi_N(z)| \leq e^{N|\Im z|}.$$

Hence, for

$$\mathcal{F}_N(z) := \varphi_N(z)\varphi_N(-z) - e^{-z^2},$$

the explicit growth bound

$$\log |\mathcal{F}_N(z)| \leq \log 2 + 4NR + 4R^2 \leq \log 2 + 4(Ba_0 + a_0^2)m, \quad |z| \leq 2R,$$

holds. On the real diameter, Lemma 3.3 gives $\log |\mathcal{F}_N| \leq -\gamma_1 m$. Apply the subharmonic two-constants theorem separately in the upper and lower half-disks of radius $2R$. The harmonic measure of the diameter is uniformly bounded below by a fixed $\omega_0 > 0$ on the concentric half-disks of radius R , so

$$\log |\mathcal{F}_N(z)| \leq -[\omega_0\gamma_1 - 4(1 - \omega_0)(Ba_0 + a_0^2)]m + O(1).$$

Choose first B large and then a_0 sufficiently small so that the bracket is a number $\gamma_2 > a_0^2$. Since

$$\inf_{|z| \leq R} |e^{-z^2}| \geq e^{-R^2} = e^{-a_0^2 m},$$

relative closeness implies $\varphi_N(z)\varphi_N(-z) \neq 0$, hence $\varphi_N(z) \neq 0$ on the disk. $\square$

3.5 Logarithmic Gaussianization and Esseen smoothing

Lemma 3.5 (Factor Gaussianization). *Under the conclusions of Lemma 3.4,*

$$d_K(X_N, \mathcal{N}(\mu_N, \sigma_N^2)) \leq \frac{C}{R},$$

where $\mu_N = \mathbb{E}X_N$ and $\sigma_N^2 = \text{Var}(X_N)$. Consequently

$$d_K(X, \mathcal{N}(0, 1)) \leq \frac{C}{\sqrt{m}}.$$

Proof. On $|z| < R$ choose the analytic logarithm

$$\psi_N(z) = \log \varphi_N(z), \quad \psi_N(0) = 0.$$

The bounded support gives $\Re \psi_N(z) = \log |\varphi_N(z)| \leq NR$. Set

$$H(z) := NR - \psi_N(z), \quad \Re H(z) \geq 0.$$

After scaling $z = Rw$, the Carathéodory coefficient inequality applied to $H(Rw)/(NR)$ gives, for $\psi_N(z) = \sum_{k \geq 1} a_k z^k$,

$$|a_k| \leq 2NR^{1-k}.$$

Since $a_1 = i\mu_N$ and $a_2 = -\sigma_N^2/2$, for every fixed $0 < \theta < 1$,

$$\left| \psi_N(t) - i\mu_N t + \frac{\sigma_N^2 t^2}{2} \right| \leq \sum_{k \geq 3} 2NR^{1-k} |t|^k \leq \frac{2N|t|^3}{R^2(1 - |t|/R)} \leq C \frac{|t|^3}{R}, \quad |t| \leq \theta R,$$

because $N/R = B/a_0$ is fixed. By Lemma 3.2, $\mu_N = o(R^{-1})$ and $\sigma_N^2 = 1 + o(R^{-1})$. Taking θ sufficiently small gives Gaussian damping, and hence

$$\left| \varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2} \right| \leq C \frac{|t|^3}{R} e^{-ct^2}.$$

The Esseen inequality in the present normalization is

$$d_K(X_N, \mathcal{N}(\mu_N, \sigma_N^2)) \leq \frac{1}{\pi} \int_{-T}^T \left| \frac{\varphi_N(t) - e^{i\mu_N t - \sigma_N^2 t^2/2}}{t} \right| dt + \frac{C}{T}.$$

With $T = \theta R$, the integral is $O(R^{-1})$ because its integrand is bounded by $Ct^2 e^{-ct^2}/R$, and the cutoff term is also $O(R^{-1})$. Add the truncation and parameter-shift errors. $\square$

Proof of Theorem 1.2. Apply Lemmas 3.1, 3.2, 3.3, 3.4, and 3.5. Since

$$m \asymp \frac{L}{\log L}, \quad L = \log(1/\Delta),$$

we have

$$\frac{1}{\sqrt{m}} \asymp \sqrt{\frac{\log L}{L}},$$

which is the claimed rate. $\square$

4 Optimality and obstruction results

4.1 A Laguerre lower-bound family

We give the lower-bound construction in full because it identifies the scale in Theorem 1.2. The only analytic input needed for the probability construction is the following uniform Laguerre envelope; its normalization and four-region proof are recorded in Appendix B.

Lemma 4.1 (Uniform Laguerre envelope).

$$C_L := \sup_{m \geq 1, x \in \mathbb{R}} \left| x e^{-x^2} L_m^{(1/2)}(3x^2/2) \right| < \infty. \quad (6)$$

Proof. The four-region estimate and the conversion from the normalized Laguerre function are given in Appendix B. $\square$

Fix $0 < \epsilon < 1/4$, let ϕ be the standard Gaussian density, and define

$$p_m(x) := x e^{-x^2} L_m^{(1/2)}(3x^2/2), \quad h_m(x) := \frac{p_m(x)}{C_L}, \quad f_m(x) := \phi(x)(1 + \epsilon h_m(x)).$$

By (6), f_m is a nonnegative probability density. It is normalized because h_m is odd. The change of variables $r = 3x^2/2$ and Laguerre orthogonality show that

$$\int_{\mathbb{R}} x^{2k+1} h_m(x) \phi(x) dx = 0, \quad 0 \leq k < m.$$

In particular, the corresponding variable U_m is centered. Its variance is one because $x^2 h_m(x) \phi(x)$ is odd. The Laguerre generating identity gives the exact MGF

$$M_{U_m}(s) = e^{s^2/2} (1 + u_m(s)), \quad u_m(s) = \frac{\epsilon}{C_L} \frac{s}{3\sqrt{3}} e^{-s^2/3} \frac{(-s^2/6)^m}{m!}. \quad (7)$$

Therefore, with

$$a_m := \frac{\epsilon}{3\sqrt{3}C_L 6^m m!},$$

we have $|u_m(s)| \leq a_m |s|^{2m+1}$ on the real axis.

Let $J_m = N_m^+ - N_m^-$, where N_m^+ and N_m^- are independent Poisson variables with mean $\lambda_m/2$, and set

$$\lambda_m := 96a_m^2 \rho^{4m-2}, \quad W_m := \frac{U_m + J_m}{\sqrt{1 + \lambda_m}}.$$

Then W_m is centered and variance one. If $q = s/\sqrt{1 + \lambda_m}$, its reflected defect is exactly

$$R_{W_m}(s) = \log(1 - u_m(q)^2) + 2\lambda_m \left(\cosh q - 1 - \frac{q^2}{2} \right). \quad (8)$$

For all sufficiently large m , $|u_m(q)| \leq 1/2$ on $|s| \leq \rho$. Using $-\log(1 - v) \leq 2v$ for $0 \leq v \leq 1/4$ and $\cosh q - 1 - q^2/2 \geq q^4/24$, the choice of λ_m gives

$$R_{W_m}(s) \geq \frac{\lambda_m q^4}{16} \geq 0.$$

The logarithmic term in (8) is nonpositive, so the upper Taylor bound gives

$$R_{W_m}(s) \leq \frac{\lambda_m \cosh(\rho) q^4}{12}.$$

If also $\lambda_m \leq 1$, evaluation at $s = \rho$ yields

$$\frac{\lambda_m \rho^4}{64} \leq \Delta_m := \Delta_\rho(W_m) \leq \frac{\lambda_m \rho^4 \cosh(\rho)}{12}.$$

Thus $\Delta_m \asymp \lambda_m$ and Stirling's formula gives

$$\log \frac{1}{\Delta_m} = 2m \log m + O(m).$$

The fixed exponential envelope is uniform: $f_m \leq (1 + \epsilon)\phi$, while

$$\mathbb{E} e^{b|J_m|} \leq 2e^{\lambda_m(\cosh b - 1)}$$

and $\sup_m \lambda_m < \infty$. Finally, the half-axis discrepancy is explicit. Since

$$\int_0^\infty h_m(x) \phi(x) dx = \frac{1}{3C_L \sqrt{2\pi}} \frac{(1/2)_m}{m!},$$

we have $|F_{U_m}(0) - 1/2| \asymp m^{-1/2}$. Moreover,

$$|F_{U_m+J_m}(0) - F_{U_m}(0)| \leq \mathbb{P}(J_m \neq 0) \leq \lambda_m = o(m^{-1/2}).$$

The normalization in W_m leaves the threshold zero unchanged. Consequently

$$d_K(W_m, \mathcal{N}(0, 1)) \geq cm^{-1/2}.$$

Together with $\log(1/\Delta_m) = 2m \log m + O(m)$, this proves Theorem 1.3 and rules out every uniform power-law modulus $C\Delta^\alpha$, $\alpha > 0$.

4.2 Failure of total-variation stability

Proposition 4.2 (Strong-distance obstruction). *There is no uniform total-variation stability theorem under the assumptions of Theorem 1.2: a standardized Poisson lattice family can satisfy a fixed exponential-integrability bound and have reflected defect tending to zero while its total-variation distance from every absolutely continuous Gaussian law equals one.*

Let $N_\lambda \sim \text{Poisson}(\lambda)$ and $X_\lambda = (N_\lambda - \lambda)/\sqrt{\lambda}$. The reflected defect satisfies, on every fixed compact s -interval,

$$R_{X_\lambda}(s) = \frac{s^4}{12\lambda} + O_\rho(\lambda^{-2}),$$

and the family has a fixed exponential-integrability bound for each fixed τ . Nevertheless X_λ is lattice-valued, whereas $\mathcal{N}(0, 1)$ is absolutely continuous, so

$$d_{\text{TV}}(X_\lambda, \mathcal{N}(0, 1)) = 1.$$

This is a qualitative boundary phenomenon and is independent of the Kolmogorov lower bound above.

5 Sample variance

Let $X_1, \dots, X_n$ be i.i.d. copies of a centered variance-one random variable X , let $\bar{X} = n^{-1} \sum_i X_i$, and set

$$Q_n = \sum_{i=1}^n (X_i - \bar{X})^2.$$

For $r \geq 1$, let V be uniform on S^{r-1} and define

$$\Psi_r(z) := \mathbb{E}_V e^{zV_1}, \quad \mathfrak{D}_{r,\tau}(Q) := \sup_{|t| \leq \tau} \left| e^{-t^2/2} \mathbb{E} \Psi_r(t\sqrt{Q}) - 1 \right|.$$

This spherical Laplace discrepancy is the natural observable for a squared radial projection; it is not a generic probability distance.

The exact sample-variance consequence of the quadratic-form theorem is the following.

Corollary 5.1 (Exact sample-variance characterization under dominance). *Let $n \geq 3$ and let $X_1, \dots, X_n$ be i.i.d., centered, and variance one. If M_X is finite near the origin,*

$$M_X(s)M_X(-s) \geq e^{s^2}$$

there, and

$$\sum_{i=1}^n (X_i - \bar{X})^2 \sim \chi_{n-1}^2,$$

then $X_i \sim \mathcal{N}(0, 1)$.

The quantitative statement is the next theorem.

Theorem 5.2 (Sample-variance stability). *Fix $n \geq 3$, $\tau, K > 0$, and $0 < \rho' < \tau\sqrt{(n-1)/n}$. There are constants $C < \infty$ and $\varepsilon_0 > 0$, depending only on (n, τ, K, ρ') , such that the*

following holds. Let $X_1, \dots, X_n$ be i.i.d. copies of a centered variance-one X with $\mathbb{E}e^{2\tau|X|} \leq K$, and assume $R_X(s) \geq 0$ for $|s| \leq \tau$. If

$$Q_n = \sum_{i=1}^n (X_i - \bar{X})^2, \quad \mathfrak{D}_{n-1, \tau}(Q_n) \leq \varepsilon \leq \varepsilon_0,$$

then

$$d_K(X, \mathcal{N}(0, 1)) \leq C \sqrt{\frac{\log \log(1/\varepsilon)}{\log(1/\varepsilon)}}.$$

We now make the discrepancy-to-defect transfer explicit. Let $r = n - 1$, let $P = I_n - n^{-1}\mathbf{1}\mathbf{1}^\top$, and choose $B \in \mathbb{R}^{r \times n}$ with $B^\top B = P$ and $BB^\top = I_r$. If V is uniform on S^{r-1} , then every column b_i of B has norm

$$q := \|b_i\| = \sqrt{\frac{n-1}{n}},$$

and $\langle b_i, V \rangle$ has the same distribution as qV_1 . Put $a = \tau q$ and define

$$D_X(s) := \frac{1}{2}R_X(s) = \frac{1}{2} \log \frac{M_X(s)M_X(-s)}{e^{s^2}}.$$

Under dominance, $D_X \geq 0$ on $[-\tau, \tau]$.

The exact spherical transform gives, for $|t| \leq \tau$,

$$e^{-t^2/2} \mathbb{E} \Psi_r(t\sqrt{Q_n}) = \mathbb{E}_V \exp \left(\sum_{i=1}^n \left[\log M_X(t\langle b_i, V \rangle) - \frac{t^2 \langle b_i, V \rangle^2}{2} \right] \right). \quad (9)$$

Jensen's inequality, exchangeability, and the symmetry of V_1 therefore imply

$$n \mathbb{E} D_X(aV_1) \leq \log(1 + \mathfrak{D}_{n-1, \tau}(Q_n)). \quad (10)$$

This is the point at which the spherical observable enters the one-dimensional reflection theory.

For the pointwise transfer, the density of V_1 is

$$f_n(v) = \frac{\Gamma((n-1)/2)}{\sqrt{\pi} \Gamma((n-2)/2)} (1-v^2)^{(n-4)/2}, \quad -1 < v < 1.$$

Fix $0 < \rho' < a$ and choose $0 < h < a - \rho'$. On $[-(\rho' + h), \rho' + h]$ the scaled density of aV_1 is bounded below by

$$m_h := \frac{1}{a} \frac{\Gamma((n-1)/2)}{\sqrt{\pi} \Gamma((n-2)/2)} \left(1 - \left(\frac{\rho' + h}{a} \right)^2 \right)^{\max(n-4, 0)/2}.$$

The exponential envelope implies that D_X is Lipschitz there. One may take

$$L_h := \frac{K}{e\tau} + \rho' + h,$$

because $M_X(s) \geq 1$ and $|x|e^{\tau|x|} \leq (e\tau)^{-1}e^{2\tau|x|}$. If $M_* := \sup_{|s| \leq \rho'} D_X(s)$, two disjoint Lipschitz tents centered at points where the maximum is attained give

$$2m_h I_{L_h, h}(M_*) \leq \frac{1}{n} \log(1 + \mathfrak{D}_{n-1, \tau}(Q_n)),$$

where

$$I_{L,h}(M) := \begin{cases} M^2/L, & M \leq Lh, \\ 2hM - Lh^2, & M > Lh. \end{cases}$$

Writing $G(y; L, h) = \sqrt{Ly}$ when $y \leq Lh^2$ and $G(y; L, h) = y/(2h) + Lh/2$ otherwise, we obtain

$$\sup_{|s| \leq \rho'} D_X(s) \leq \inf_{0 < h < a - \rho'} G\left(\frac{\log(1 + \mathfrak{D}_{n-1,\tau}(Q_n))}{2nm_h}; Lh, h\right). \quad (11)$$

In particular, for fixed n, τ, K, ρ' , the right-hand side is $O(\sqrt{\mathfrak{D}_{n-1,\tau}(Q_n)})$ as the discrepancy tends to zero. The strict interior restriction is essential when $n \geq 5$, because the spherical density vanishes at the endpoints.

Applying (11) to the sample-variance hypothesis and then Theorem 1.2 proves Theorem 5.2. In the exact case, Corollary 5.1 gives Gaussianity under the same dominance condition. Thus the discrepancy-to-defect argument is a quantitative extension of the exact spherical-Jensen mechanism.

Two questions remain open. Can fixed exponential integrability be replaced by a sharp tail or moment hypothesis without changing the reflection modulus? Under which additional uniform-integrability hypothesis can a standard distributional distance replace $\mathfrak{D}_{r,\tau}$ while retaining the same interior pointwise transfer? A further direction is to extend the factor-recovery and optimality theory from the i.i.d. sample-variance projection to general positive-semidefinite quadratic forms with non-identically distributed coordinates.

A Parameter selection for the sharp upper bound

This appendix records a non-circular order of choices for the five-step proof. Fix (ρ, τ, K) and an interior analytic radius. Let $\omega_0 > 0$ be the lower bound for the harmonic measure of the real diameter in the smaller half-disk used in the propagation step, and let C_G be the corresponding complex-growth constant. Choose $A > 8$ and then choose $c_0 > 0$ so small that

$$(A + 3)c_0 < \omega_0.$$

With $m = \lfloor c_0 L / \log L \rfloor$, this makes the fixed-disk Cauchy estimate dominate all factorial losses in the moment-transfer lemma.

Next choose the truncation factor B sufficiently large that the square-root tail estimate has exponent $\beta_B > 4$. After B is fixed, choose $a_0 > 0$ so small that the real Taylor remainder obeys $C_T a_0^2 < e^{-4}$ and, with $\gamma_1 = -\log(C_T a_0^2)$,

$$C_G(Ba_0 + a_0^2) < \frac{\omega_0 \gamma_1}{2}, \quad \gamma_2 := \omega_0 \gamma_1 - C_G(Ba_0 + a_0^2) > a_0^2.$$

Such an a_0 exists because the left-hand growth term tends to zero while γ_1 tends to infinity as $a_0 \downarrow 0$. Finally choose the fixed Esseen fraction $0 < \theta < 1/4$ so that the cubic remainder in the logarithmic expansion is at most one quarter of the Gaussian quadratic damping on $|t| \leq \theta R$. All constants then depend only on the fixed model parameters and not on Δ or m .

B A uniform Laguerre estimate

For $\alpha = 1/2$, use the normalization

$$\mathcal{L}_m^{(\alpha)}(r) = a_{m,\alpha} L_m^{(\alpha)}(r) e^{-r/2} r^{\alpha/2}, \quad a_{m,\alpha} = \sqrt{\frac{m!}{\Gamma(m + \alpha + 1)}}, \quad \nu = 4m + 2\alpha + 2.$$

Proposition 3.2 of [9] gives

$$\begin{aligned} |\mathcal{L}_m^{(1/2)}(r)| &\leq C(r\nu)^{1/4} \quad (0 \leq r \leq \nu^{-1}), \\ |\mathcal{L}_m^{(1/2)}(r)| &\leq C(r\nu)^{-1/4} \quad (\nu^{-1} \leq r \leq \nu/2), \\ |\mathcal{L}_m^{(1/2)}(r)| &\leq C\nu^{-1/4}(\nu^{1/3} + |\nu - r|)^{-1/4} \quad (\nu/2 \leq r \leq 3\nu/2), \end{aligned}$$

and an exponential estimate for $r \geq 3\nu/2$. Since

$$\sqrt{r} e^{-r/2} |L_m^{(1/2)}(r)| = \sqrt{\frac{\Gamma(m + 3/2)}{m!}} r^{1/4} |\mathcal{L}_m^{(1/2)}(r)|,$$

the first two regions are uniformly bounded after multiplication by $e^{-r/6}$. In the turning region the same multiplication contributes $e^{-r/6} \leq e^{-\nu/12}$, which dominates the polynomial factor from the displayed estimate, and the outer estimate is exponentially decaying. Consequently,

$$\sup_{m \geq 1, r \geq 0} \sqrt{r} e^{-2r/3} |L_m^{(1/2)}(r)| < \infty.$$

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